

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.4 Worksheet

1. Find the cross product $\mathbf{a} \times \mathbf{b}$ and verify that it is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

a) $\mathbf{a} = \langle 6, 0, -2 \rangle$ & $\mathbf{b} = \langle 0, 8, 0 \rangle$

b) $\mathbf{a} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ & $\mathbf{b} = -\mathbf{i} + 5\mathbf{k}$

c) $\mathbf{a} = \left\langle t, 1, \frac{1}{t} \right\rangle$ & $\mathbf{b} = \langle t^2, t^2, 1 \rangle$

2. State whether each expression is defined or not. If not, explain why. If so, state whether it is a vector or a scalar.

a) $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$

b) $\mathbf{a} \times (\mathbf{b} \cdot \mathbf{c})$

c) $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$

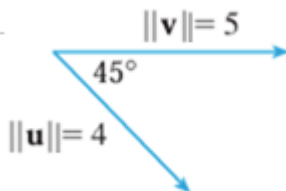
d) $\mathbf{a} \cdot (\mathbf{b} \cdot \mathbf{c})$

e) $(\mathbf{a} \cdot \mathbf{b}) \times (\mathbf{c} \cdot \mathbf{d})$

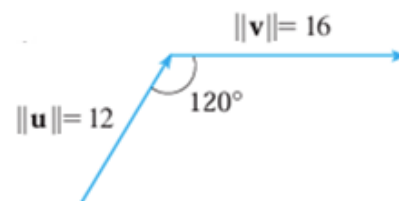
f) $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d})$

3. Find $\|\mathbf{u} \times \mathbf{v}\|$ and determine whether $\mathbf{u} \times \mathbf{v}$ is directed into the page or out of the page.

a)

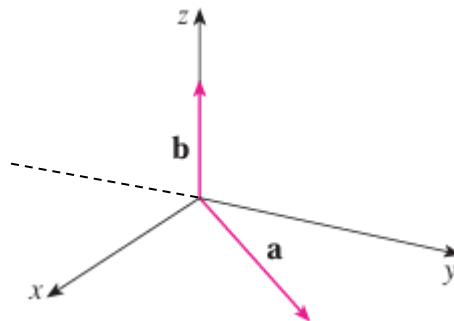


b)



4. The figure shows a vector $\mathbf{a}$ in the xy -plane and a vector $\mathbf{b}$ in the direction of $\mathbf{k}$. Their lengths are $\|\mathbf{a}\| = 3$ and $\|\mathbf{b}\| = 2$.

a) Find $\|\mathbf{a} \times \mathbf{b}\|$.



b) Use the right-hand rule to decide whether the components of $\mathbf{a} \times \mathbf{b}$ are positive, negative or 0.

5. Consider the vectors $\mathbf{a} = \langle 1, 2, 3 \rangle$ & $\mathbf{b} = \langle -1, 1, 2 \rangle$

a) Find the area of the parallelogram determined by these vectors.

b) Find the area of the triangle determined by these vectors.

c) Determine if $\mathbf{a}$ and $\mathbf{b}$ are coplanar with the vector $\mathbf{c} = \langle 2, 1, 4 \rangle$.

6. Consider the vectors $\mathbf{a} = \mathbf{i} + \mathbf{j}$ & $\mathbf{b} = \mathbf{j} + \mathbf{k}$

a) Find the area of the parallelogram determined by these vectors.

b) Find the area of the triangle determined by these vectors.

c) Determine if $\mathbf{a}$ and $\mathbf{b}$ are coplanar with the vector $\mathbf{c} = \mathbf{i} + \mathbf{j} - \mathbf{k}$.

7. If $\mathbf{a} \cdot \mathbf{b} = \sqrt{3}$ and $\mathbf{a} \times \mathbf{b} = \langle 1, 2, 2 \rangle$, find the angle between $\mathbf{a}$ and $\mathbf{b}$.

8. Suppose vector $\mathbf{v}$ is a vector in $\mathbb{R}^3$.

a) Find all vectors $\mathbf{v}$ such that $\langle 1, 2, 1 \rangle \times \mathbf{v} = \langle 3, 1, -5 \rangle$.

b) Explain why there is no vector $\mathbf{v}$ such that $\langle 1, 2, 1 \rangle \times \mathbf{v} = \langle 3, 1, 5 \rangle$.

9. Let P be a point not on the line L that passes through the points Q and R . Show that the distance from point P to the line L is $d = \frac{\|\mathbf{a} \times \mathbf{b}\|}{\|\mathbf{a}\|}$, where $\mathbf{a} = \overrightarrow{QR}$ and $\mathbf{b} = \overrightarrow{QP}$.

10. If $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$, show that $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a}$. Use properties of the cross product to help show that $\mathbf{a} \times \mathbf{b}$ is equal to both $\mathbf{b} \times \mathbf{c}$ and then that $\mathbf{b} \times \mathbf{c}$ is equal to $\mathbf{c} \times \mathbf{a}$.

11. Prove that $(\mathbf{a} - \mathbf{b}) \times (\mathbf{a} + \mathbf{b}) = 2(\mathbf{a} \times \mathbf{b})$.

12. Suppose that $\mathbf{a} \neq \mathbf{0}$.

a) If $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$, does it follow that $\mathbf{b} = \mathbf{c}$?

b) If $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$, does it follow that $\mathbf{b} = \mathbf{c}$?

c) If $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$ and $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$, does it follow that $\mathbf{b} = \mathbf{c}$?